

IBDC10050_CAN using examples

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日期：2023.4.3

Example 1: Read the low voltage side battery voltage of module 4

direction	CAN ID	Data type	types of fame	DLC	Data
Send	0x0074530F	Data frame	Extended Frame	0x08	00 00 00 04 00 00 00 00
Receive	0x00F20F53	Data frame	Extended Frame	0x08	01 BC 00 04 FF FF 00 00

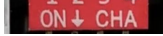
Send: CAN ID: 0x0074530F

analysis:

0x74: The type of frame command is: read command (0x73 is for setting module information; 0x74 is for reading module information)

0x53: Target address (0x50 is the No. 1 module, 0x51 is the No. 2 module, 0x52 is the No. 3 module, 0x53 is the No. 4 unit, and so on. Note that the module number is determined by the module's Dip switch status)



For example, the status of the Dip switch: , Indicates that the module address dial status is: 0000 (binary)=0 (decimal), which is module 1, and the target address is: 0x50.

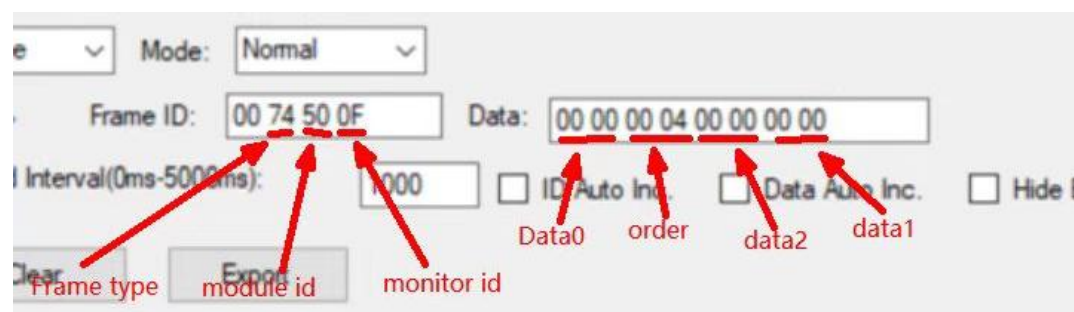


For example, the status of the Dip switch: , Indicates that the module address dial status is: 0001 (binary)=1 (decimal), which is the module number 2. The target address is: 0x51.

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

CAN Data: 00 00 00 04 00 00 00 00

Byte order	Data0	Order	Data2	Data1
Byte Value	00 00	00 04	00 00	00 00



analysis: **Order** = 4 is:

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter			
Order	Data0	Data1	Data2
The following data is defined as DCDC unit data			
3	Battery charging state 0 --- Standby, 1 --- Constant current, 2 --- Constant power, 3 --- Constant voltage, 4 --- Float charge, 5 --- Full charge	Charging power unit: W	Discharge power unit: W
4	Battery voltage unit: V	Battery current unit: A	RSVD

Receive:

CAN ID: 0x00F20F53

analysis:

0xF2: The frame type is: Set Response (0x72/0xF2: Query or Set Response Frame)

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x53: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

CAN Data:

01 BC

00 04

FF FF

00 00

analysis:

Battery Voltage

Order

RSVD

Battery Current

Battery pack voltage is: $44.4V = 0x1BC * 0.1(\text{fact}) = 444 * 0.1$, current is: 0.0A

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter			
Order	Data0	Data1	Data2
The following data is defined as DCDC unit data			
3	Battery charging state	Charging power unit: W	Discharge power unit: W
	0 -- Standby,		
	1 -- Constant current,		
	2 -- Constant power,		
	3 -- Constant voltage,		
	4 -- Float charge,		
	5 -- Full charge		
4	Battery voltage unit: V	Battery current unit: A	RSVD

Example 2: Read the bus voltage on the high voltage side of module 4

direction	Frame ID	Data type	Frame type	DLC	Data
Send	0x0073530F	Data frame	Extended Frame	0x08	00 00 00 06 00 00 00 00
Received	0x00F20F53	Data frame	Extended Frame	0x08	00 00 00 06 00 00 08 90

Send:

CAN ID: 0x0074530F

analysis:

0x74: The type of frame command is: read command (0x73 is for setting module information; 0x74 is for reading module information)

0x53: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data: 00 00 00 06 00 00 00 00

analysis: Order = 6 Bus voltage return

Received:

CAN ID: 0x00F20F53

analysis:

0xF2: Frame type: query response (0x72/0xF2: query or set response frame)

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x53: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Data: 00 00 00 06 00 00 08 90
analysis: RSVD Order RSVD BUS Voltage

BUS voltage is: $219.2V = 0x890 * 0.1(\text{fact}) = 2192 * 0.1 = 219.2v$

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter			
Order	Data0	Data1	Data2
The following data is defined as DCDC unit data			
4	Battery voltage unit: V	Battery current unit: A	RSVD
5	RSVD	RSVD	RSVD
6	RSVD	BUS bus voltage unit: V	RSVD
7	RSVD	RSVD	RSVD

Example 3:

Set up a high-voltage bus to charge the low-voltage battery pack e.g.

Finish: Step1 to Step4

Step1: Set DCDC as the charging mode from the high-voltage bus to low-voltage battery pack

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 01 00 34 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

This message is: Set the module 2, Order=52(charge and discharge mode);

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data: 00 01 00 34 00 00 00 00
analysis: Data0 Order Data2 Data1

Order = 0x34 = 52 Indicates the type of charge and discharge command

Data0 = 0x0001 = 1 Indicates charging

A	B	C	D	E	F	G
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
1 - Lithium battery						
49 RSVD		RSVD	RSVD	USHORT	1	R
DC bus voltage unit: V						
50 data range [23004250]; Default value: 3800		RSVD	RSVD	USHORT	0.1	R/W
Nominal battery capacity						
51 data range [381000]; Default value: 100		RSVD	RSVD	USHORT	1	R/W
Charging and discharging command						
0 --- Standby						
1 --- Charging						
2 --- Discharge						
3 --- BAT automatic switching						
4 --- BBS automatic switching						
5 --- Off charging						
52		RSVD	RSVD	USHORT	1	R/W
Charging and discharging power setting						
53 data range [0120] Unit:W Default value: 100W		RSVD	RSVD	USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step2: Set the charging voltage value of the low-voltage battery pack

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	02 58 00 36 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis:

02 58	00 36	00 00	00 00
Data0	Order	Data2	Data1

Order = 0x36 = 54 Indicates the command to set the charging voltage

Data0 = 0x0258 = 600 (Unit:V, Fact:0.1, Offset:0, 600*0.1 = 60V) battery pack
Charging Voltage value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
53	Charging and discharging power setting data range [0120] Unit: % Default value: 100%	RSVD	RSVD	USHORT	1	R/W
54	Battery balance charging voltage data range [1001100] Unit: V Default value: 600	RSVD	RSVD	USHORT	0.1	R/W
55	Charging & discharging current setting value range [01300] Unit: A Default value: 1040	RSVD	RSVD	USHORT	0.1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step3: Set the charging current value of the low-voltage battery pack

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	01 F4 00 37 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis

01 F4	00 37	00 00	00 00
Data0	Order	Data2	Data1

Order = 0x37 = 55, Indicates the command to set the charging and discharging current, which refers to the charging here

Data0 = 0x01F4 = 500 (Unit:A, Fact:0.1, Offset:0, 500*0.1 = 50A) battery pack Charging current value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
53	Charging and discharging power setting data range [0120] Unit: % Default value: 100%	RSVD	RSVD	USHORT	1	R/W
54	Battery balance charging voltage data range [1001100] Unit: V Default value: 600	RSVD	RSVD	USHORT	0.1	R/W
55	Charging & discharging current setting value range [01300] Unit: A Default value: 1040	RSVD	RSVD	USHORT	0.1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step4: Set the power module to power on and start

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 55 00 20 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

Analysis:



Order = 0x20 = 32, Indicates the command to set the module ON/OFF

Data0 = 0x0055 (0x0055: Module startup, 0x00AA:Module shutdown)

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter							
Order		Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data							
29	Fault clearing 0x0055 - Clear all faults	RSVD	RSVD		USHORT	1	R/W
30	Operation mode setting 0: No output mode 1: Normal operation mode 2: Open loop debugging mode 3: Wave generation debugging mode	RSVD	RSVD		USHORT	1	R/W
31	Operation status setting 0: Adaptive 1: Grid connected 2: Off grid	RSVD	RSVD		USHORT	1	R/W
32	System On/Off 0x0055 -- On 0x00AA -- Off	RSVD	RSVD		USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Example 4:

Set up a Low-voltage battery pack to discharge the high-voltage bus e.g.

Finish: Step1 to Step4

Step1: Set DCDC as the discharging mode from the low-voltage battery pack to high-voltage bus

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 02 00 34 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

This message is: Read and set the 2nd module Order=52, charge and discharge mode;

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data: 00 02 00 34 00 00 00 00
analysis: Data0 Order Data2 Data1

Order = 0x34 = 52 Indicates the type of charge and discharge command

Data0 = 0x0002 = 2 Indicates discharging

A	B	C	D	E	F	G
CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
1 - Lithium battery						
49	RSVD	RSVD	RSVD	USHORT	1	R
50	DC bus voltage unit: V data range [23004250]; Default value: 3800	RSVD	RSVD	USHORT	0.1	R/W
51	Nominal battery capacity data range [381000]; Default value: 100	RSVD	RSVD	USHORT	1	R/W
52	Charging and discharging command 0 --- Standby 1 --- Charging 2 --- Discharge 3 --- BAT automatic switching 4 --- BBS automatic switching 5 --- Off charging	RSVD	RSVD	USHORT	1	R/W
53	Charging and discharging power setting data range [0120] Unit: % Default value: 100%	RSVD	RSVD	USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step2: Set the discharge voltage value from the low-voltage battery pack discharge to the high-voltage bus

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	0E D8 00 32 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis:

0E D8	00 32	00 00	00 00
Data0	Order	Data2	Data1

Order = 0x32 = 50 Indicates the command to set the discharge voltage from low voltage to high voltage bus

Data0 = 0x0ED8 = 3800 (Unit:V, Fact:0.1, Offset:0, 3800*0.1=380V) High voltage bus DisCharging Voltage value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter				
Order	Data0	Data1	Data2	
The following data is defined as DCDC unit data				
44	Serial Number 7	RSVD	RSVD	
45	Serial Number 8	RSVD	RSVD	
46	Serial Number 9	RSVD	RSVD	
47	Serial Number 10	RSVD	RSVD	
48	Battery type 0 - Lead acid battery 1 - Lithium battery	RSVD	RSVD	
49	RSVD	RSVD	RSVD	
50	DC bus voltage unit: V data range [23004250]; Default value: 3800	RSVD	RSVD	
51	Nominal battery capacity data range [381000]; Default value: 100	RSVD	RSVD	

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step3: Set the discharge current value on the high-voltage busbar side

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	01 F4 00 37 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis: Data0

01 F4	00 37	00 00	00 00
Data0	Order	Data2	Data1

Order = 0x37 = 55 Indicates the command to set the charging and discharging current, which refers to the discharging current here

Data0 = 0x01F4 = 500 (Unit:A, Fact:0.1, Offset:0, 500*0.1 = 50A) battery pack Charging current value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
53	Charging and discharging power setting data range [0:20] Unit: % Default value: 100%	RSVD	RSVD	USHORT	1	R/W
54	Battery balance charging voltage data range [100:1100] Unit: V Default value: 600	RSVD	RSVD	USHORT	0.1	R/W
55	Charging & discharging current setting value range [0:1300] Unit: A Default value: 1040	RSVD	RSVD	USHORT	0.1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step4: Set the power module to power on and start

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 55 00 20 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Set power module startup frame type: Set module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis: Data0



Order = 0x20 = 32 Indicates the command to set the module ON/OFF

Data0 = 0x0055 (0x0055: Module startup, 0x00AA:Module shutdown)

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter							
Order		Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data							
29	Fault clearing 0x0055 - Clear all faults	RSVD	RSVD		USHORT	1	R/W
30	Operation mode setting 0: No output mode 1: Normal operation mode 2: Open loop debugging mode 3: Wave generation debugging mode	RSVD	RSVD		USHORT	1	R/W
31	Operation status setting 0: Adaptive 1: Grid connected 2: Off grid	RSVD	RSVD		USHORT	1	R/W
32	System On/Off 0x0055 -- On 0x00AA -- Off	RSVD	RSVD		USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Example 5: Set module grouping number.

Finish: Step1 to Step2

Step1: Set module group number

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 02 00 22 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Message Analysis: Order=34, set module 2 grouping, group module 2 to group 2
Send:

CAN ID: 0x0073510F

Analysis:

0x73: Frame type: set module (0x73:set module information; 0x74: read module information)

0x51: Destination address (0x50 for module 1, 0x51 for module 2, 0x53 for module 4, and so on)

0x0F: (group ID = 1)<<3 + source address (group ID is 1 and source address is 0x7)

Data:	00 02	00 22	00 00	00 00
Analysis:	Data0	Order	Data2	Data1

Order = 0x22 = 34 Indicates the command type: set module grouping command

Data0 = 0x0002 = 2 Indicates setting the module to group 2

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter				
Order	Data0	Data1	Data2	
The following data is defined as DCDC unit data				
33	Restore factory settings 0x0055 -- Restore all values 0x00AA -- Restore user values	RSVD	RSVD	
34	Group ID number data range [1100]	RSVD	RSVD	

Received: CAN ID: 0x00220F51

Step2: Reboot Power module

Note:

If you want to operate Module 2 after setting up the module group successfully and rebooting, the CAN ID needs to be adjusted according to the Group ID of Module 2.

For example, to set the operation of module 2: CAN ID = 0x00735117

Analysis:

0x73: Frame type: set module (0x73 to set module information; 0x74 to read module information)

0x17: (group ID = 2)<<3 + source address (new group ID is 2, default source address is 0x7)

0x51: Destination address (0x50 for module 1, 0x51 for module 2, 0x5F is intra-group broadcast)

For example: To read operation on module 2 : CAN ID = 0x00745117

Analysis:

0x74: Frame type: read module (0x73 to set module information; 0x74 to read module information)

0x17: (group ID = 2)<<3 + source address (new group ID is 2, default source address is 0x7)

0x51: Destination address (0x50 is module 1, 0x51 is module 2, 0x5F is intra-group broadcast)

Example 6:

Set up Set whether to charge or discharge the low-voltage battery pack from the high-voltage bus, to maintain the voltage of the high-voltage bus unchanged, that is, to set the automatic switching of the high-voltage bus e.g.

Finish: Step1 to Step4

Step1: Set DCDC as the automatic switching of the high-voltage bus

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 04 00 34 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

This message is: Read and set the 2nd module Order=52, charge and discharge mode;

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data: 00 04 00 34 00 00 00 00
analysis: Data0 Order Data2 Data1

Order = 0x34 = 52 Indicates the type of charge and discharge command

Data0 = 0x0004 = 4 Indicates Bus automatic switching

A	B	C	D	E	F	G
CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
1 - Lithium battery						
49	RSVD	RSVD	RSVD	USHORT	1	R
50	DC bus voltage unit: V data range [23004250]; Default value: 3800	RSVD	RSVD	USHORT	0.1	R/W
51	Nominal battery capacity data range [381000]; Default value: 100	RSVD	RSVD	USHORT	1	R/W
52	Charging and discharging command 0 --- Standby 1 --- Charging 2 --- Discharge 3 --- BAT automatic switching 4 --- BUS automatic switching 5 --- Off charging	RSVD	RSVD	USHORT	1	R/W
53	Charging and discharging power setting data range [0120] Unit:W Default value: 100W	RSVD	RSVD	USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step2: Set the the high-voltage bus voltage

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	0E D8 00 32 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis:

0E D8	00 32	00 00	00 00
Data0	Order	Data2	Data1

Order = 0x32 = 50 Indicates the command to set the discharge voltage from low voltage to high voltage bus

Data0 = 0x0ED8 = 3800 (Unit:V, Fact:0.1, Offset:0, 3800*0.1=380V) High voltage bus Voltage value(380V is an example voltage)

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter			
Order	Data0	Data1	Data2
The following data is defined as DCDC unit data			
44	Serial Number 7	RSVD	RSVD
45	Serial Number 8	RSVD	RSVD
46	Serial Number 9	RSVD	RSVD
47	Serial Number 10	RSVD	RSVD
48	Battery type 0 - Lead acid battery 1 - Lithium battery	RSVD	RSVD
49	RSVD	RSVD	RSVD
50	DC bus voltage unit: V data range [23004250]; Default value: 3800	RSVD	RSVD
51	Nominal battery capacity data range [381000]; Default value: 100	RSVD	RSVD

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step3: Set the discharge and charge Power value

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 64 00 35 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:
CAN ID: 0x0073510F
analysis:
0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)
0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)
0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data: 00 64 00 35 00 00 00 00
analysis: Data0 Order Data2 Data1

Order = 0x35 = 53 Indicates the command to set the charging and discharging power,
Data0 = 0x0064 = 100 (Unit:%, Fact:1, Offset:0, 100*1 = 100%) charging and discharging power value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
53	Charging and discharging power setting data range [0120] Unit:% Default value: 100%	RSVD	RSVD	USHORT	1	R/W
54	Battery balance charging voltage data range [1001100] Unit: V Default value: 600	RSVD	RSVD	USHORT	0.1	R/W
55	Charging & discharging current setting value range [01300] Unit: A Default value: 1040	RSVD	RSVD	USHORT	0.1	R/W

Received:
CAN ID: 0x00220F51
analysis:
0x22: Frame Type: Set Frame Response
0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)
0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step4: Set the discharge and charge current value

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	01 F4 00 37 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Frame type: Setting module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis: Data0

01 F4

00 37

00 00

00 00

Order

Data2

Data1

Order = 0x37 = 55 Indicates the command to set the charging and discharging current, which refers to the discharging current here

Data0 = 0x01F4 = 500 (Unit:A, Fact:0.1, Offset:0, $500 \times 0.1 = 50A$) battery pack Charging current value

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
53	Charging and discharging power setting data range [0120] Unit: % Default value: 100%	RSVD	RSVD	USHORT	1	R/W
54	Battery balance charging voltage data range [1001100] Unit: V Default value: 600	RSVD	RSVD	USHORT	0.1	R/W
55	Charging & discharging current setting value range [01300] Unit: A Default value: 1040	RSVD	RSVD	USHORT	0.1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

Step5: Set the power module to power on and start

direction	CAN ID	Data type	Frame type	DLC	Data
Send	0x0073510F	Data frame	Extended Frame	0x08	00 55 00 20 00 00 00 00
Received	0x00220F51	Data frame	Extended Frame	0x08	

Send:

CAN ID: 0x0073510F

analysis:

0x73: Set power module startup frame type: Set module (0x73 for setting module information; 0x74 for reading module information)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)

0x0F: Group ID<<3+Source Address (Group ID is 1, Source Address is 0x7)

Data:

analysis: Data0

00 55

00 20

00 00

00 00

Order

Data2

Data1

Order = 0x20 = 32 Indicates the command to set the module ON/OFF

Data0 = 0x0055 (0x0055: Module startup, 0x00AA:Module shutdown)

CAN Bus Protocol Data Sheet for Bidirectional DCDC Converter						
Order	Data0	Data1	Data2	data type	Fact	attribute
The following data is defined as DCDC unit data						
29	Fault clearing 0x0055 - Clear all faults	RSVD	RSVD	USHORT	1	R/W
30	Operation mode setting 0: No output mode 1: Normal operation mode 2: Open loop debugging mode 3: Wave generation debugging mode	RSVD	RSVD	USHORT	1	R/W
31	Operation status setting 0: Adaptive 1: Grid connected 2: Off grid	RSVD	RSVD	USHORT	1	R/W
32	System On/Off 0x0055 - On 0x00AA - Off	RSVD	RSVD	USHORT	1	R/W

Received:

CAN ID: 0x00220F51

analysis:

0x22: Frame Type: Set Frame Response

0x0F: Group ID<<3+Source Address (default group ID is 1, default source address is 0x7)

0x51: Target address (0x50 for Unit 1, 0x51 for Unit 2, 0x53 for Unit 4, and so on)